Algebraic Frequency Grids, Harmonic Resonances and Primality (vf4)

A Lean 4 / Mathlib formalization of the unconditional part

`FrequencyGrids` Lean development

Lean 4.33.1 (`leanprover/lean4:v4.33.1`) with Mathlib

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Abstract

This document reports a formalization, in Lean 4 with Mathlib, of the six unconditional claims of *Algebraic Frequency Grids, Harmonic Resonances and Primality* (vf4): the existence of the degree-6 root $s_2(t)$ for the characteristic equation, the exact lattice step $b(t,k{+}1,\sigma){-}b(t,k,\sigma) = -2\pi/\ln 2$ (hence $N_{\text{phase}} = 0$), the $O(T)$ counting of the resonant stratum with the observed slope $200\ln 2/\pi \approx 44.127$ and the distinct-versus-total dichotomy $N_R \le N_{\text{tot}}$ (equality iff no collisions), the off-diagonal Gram bound $|G_{ij}(T)| \le 2/(T|b_i - b_j|)$, the invariance of $\operatorname{Re} S_f(N,b)$ under $b \mapsto -b$, and the $6n\pm1$ filter with its failing converse. The conjectural claims ($\eta_R(T) \to 1$, $N_{\text{off}} = o(N_R)$, $N_{\text{van}} = o(N_R)$, and the primality criterion $\operatorname{Re}(S_f(n-1)S_f(n+1)) < 0$) are *stated only*, each as a `def ... : Prop` with no proof; nothing in the development assumes them. The two diagnostics they mention, N_{off} and N_{van}, are the paper's own concrete objects (§9): the off-diagonal Gram interference and the count of clamped eigenvalues of the Gram matrix of the resonant modes below the relative threshold δ_T. Implementing them faithfully has a consequence worth stating up front: at the paper's own parameters those two counts are *not* small compared with N_R (§9.4), so the experiments of vf4 §6.4.8 do not support the conjectures they are meant to support. Making the diagnostics concrete also settles what the conjecture *demands*: $G = EE^{\mathsf{H}}$ is positive semidefinite of rank at most X, so at least $\#\text{modeSet}(T) - X$ modes have zero eigenvalue and are counted by N_{van}. Hence $N_{\text{van}} = o(N_R)$ forces $X(T) \ge (1-\varepsilon)N_{\text{tot}}(T)$ for every $\varepsilon > 0$, and any truncation with $X \le c\,N_R$, $c < 1$, refutes the conjecture (§10). The factorisation is proved for an *arbitrary* frequency family $f : \iota \to \mathbb{R}$ (§9.2), which yields two instances: the triplet-indexed matrix of the construction, and the matrix indexed by the *distinct* resonant frequencies, whose index type has cardinality $N_R(T)$. For the latter the bound $N_{\text{van}} \ge N_R - \lfloor X \rfloor$ holds unconditionally, so the refutation needs no hypothesis on collisions.

Two discrepancies with the paper's numerics are proved or checked and reported in §3 and §5: (i) the bracketing interval $(-0.9999, -0.0001)$ carries a sign change only for $1 \le t \le 589$ and loses it from $t = 590$ onward — the root exists for all $t \ge 1$ but leaves the interval; (ii) the value $s_2 = -0.99941091795$ hard-coded in vf4 §7.3–7.4 is the root at $t = 100$, not at $t = 1$. The development is `sorry`-free and depends only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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1 Scope and result

What is formalized

`FrequencyGrids` is a self-contained Lean 4 development on top of Mathlib. It has no project-specific axioms and no `sorry`. The six items of the task map onto the files as follows.

- **Item 1** (`FrequencyGrids/Basic.lean`). The characteristic equation

$$E(s, k_3) = \left|8s^6 - 24s^4 + 24s^2 - 8\right|^{2/3}(24k_3^2 s^6 - 72k_3^2 s^4 - 24k_3^2) + 4s^4$$

  is shown to be a *polynomial* in s (the fractional power cancels a perfect cube), whence its continuity and a real root $s_2(t) \in (-1, 0)$ of $E(\cdot, -t)$ for every $t \geq 1$; the narrow interval $(-0.9999, -0.0001)$ of the paper's `brentq` call is analysed exactly, with the sign change verified for $1 \leq t \leq 589$ and proved to fail for $t \geq 590$.

- **Item 2** (`FrequencyGrids/Lattice.lean`). The b-lattice has constant step $-2\pi/\ln 2$ independent of t, k, σ, the map $k \mapsto b(t, k, \sigma)$ is strictly antitone (hence injective), and the phase defect vanishes identically, so that $N_{\mathrm{phase}} = 0$ for *every* truncation.

- **Item 3** (`FrequencyGrids/Counting.lean`, `FrequencyGrids/Structure.lean`). A finite set of integers in a real interval $[a, b]$ has at most $b - a + 1$ elements and, if it contains all of $[a, b] \cap \mathbb{Z}$, at least $b - a - 2$; this gives $\left|\#\{k \neq 0 : |b(t, k, \sigma)| \leq T\} - T \ln 2/\pi\right| \leq 3$ for each of the 200 lattices and hence $\left|N_{\mathrm{tot}}(T) - 200\, T \ln 2/\pi\right| \leq 600$. The distinct count satisfies $N_R \leq N_{\mathrm{tot}}$, with equality exactly when there is no collision, and collisions themselves are rigid: they force equal sign, equal index and equal $|s_2|$.
- **Item 4** (`FrequencyGrids/Resonance.lean`). The Gram coefficient $G_{12}(T)$ is defined as in vf4 §6.5.2.1 and shown to satisfy $|G_{12}(T)| \leq 2/(T|b_1 - b_2|)$ for $b_1 \neq b_2$, $T > 0$.
- **Item 5** (`FrequencyGrids/Resonance.lean`). $\overline{S_f(N, b)} = S_f(N, -b)$ for the partial harmonic sums $S_f(N, b) = \sum_{m \leq N} m^{-(1/2+ib)}$, hence $\operatorname{Re} S_f(N, \cdot)$ is even.
- **Item 6** (`FrequencyGrids/Congruence.lean`). Every prime $p > 3$ is $\equiv \pm 1 \pmod 6$, the converse fails at 25, and the filter is therefore not a primality test.
- **The diagnostics** (`FrequencyGrids/Eigen.lean`). The Gram matrix of the resonant modes, its spectrum, the relative interference r_j and the counts N_{off}, N_{van} are the paper's own objects; $G = EE^{\mathsf{H}}$ is proved, whence positive semidefiniteness, nonnegative eigenvalues, the identity $\max(\lambda, 0) = \lambda$, the rank bound $\operatorname{rank} G \leq \operatorname{rank} E \leq \lfloor X \rfloor$, the diagonal $G_{ii} = H_{\lfloor X \rfloor}$, and $N_{\mathrm{van}} \geq N_{\mathrm{tot}} - \lfloor X \rfloor$ (§9.1).
- **The general family** (`FrequencyGrids/GeneralGram.lean`). The same is proved for an arbitrary finite frequency family $f : \iota \to \mathbb{R}$, with E indexed either by $\operatorname{Fin} \lfloor X \rfloor$ (real cutoff) or by $\operatorname{Fin} X$ (natural cutoff). Instantiating at the *distinct* resonant frequencies — an index type of cardinality $N_R(T)$ — gives $N_{\mathrm{van}}(F) \geq N_R - \lfloor X \rfloor$ with no collision hypothesis (§9.2).
- **The truncation** (`FrequencyGrids/Necessary.lean`). With X a parameter of the conjecture, $N_{\mathrm{van}} = o(N_R)$ implies $X(T) \geq (1 - \varepsilon) N_{\mathrm{tot}}(T)$ eventually for every $\varepsilon > 0$, so a truncation with $X \leq c\, N_R$, $c < 1$, refutes it (§10).

What is not formalized, and why

- The conjectures $\eta_R \to 1$, $N_{\mathrm{off}} = o(N_R)$, $N_{\mathrm{van}} = o(N_R)$ and the primality criterion are `def ... : Prop` in `FrequencyGrids/Conjectures.lean`. No theorem in the development mentions them; `#print axioms` on those declarations returns nothing but the three standard axioms, because there is nothing to check.
- N_{off} and N_{van} are now the paper's own diagnostics (§9): the Gram matrix of the resonant modes, its eigenvalues via the spectral theorem for Hermitian matrices, the relative interference $r_j(X)$ and the relative eigenvalue threshold $\delta_T \max_j \lambda_j$. The one idealisation left is numerical: the eigenvalues are exact real numbers (Mathlib's spectral theorem) where the paper uses floating point. The factorisation $G = EE^{\mathsf{H}}$, the positive semidefiniteness, the identity $\max(\lambda_j, 0) = \lambda_j$ and the rank bound $\operatorname{rank} G \leq X$ are all proved (§9.1).
- The numerical statement $N_{\mathrm{off}} = \#\mathrm{modeSet}$ at the parameters of vf4 §6.4.8 is *not* proved. The reduction to what would suffice is formalized (`Noff_eq_card_of_gram_gt`); §9.4 reports the two obstructions.
- The integral representation $G_{ij}(T) = \frac{1}{T} \int_0^T e^{i(b_i - b_j)t}\, dt$ is quoted in the paper as the *definition* of G_{ij}; we take its closed form as the definition and prove the bound. The equality of the two was not formalized.
- The deduction "$N_{\mathrm{phase}}, N_{\mathrm{off}}, N_{\mathrm{van}} = o(N_R) \Rightarrow \eta_R \to 1$" is elementary but is not formalized: it would be a theorem about the conjectures' *consequences*, and stating it without the conjectures risks reading as a proof of them.
- Uniqueness of the root $s_2(t)$, and injectivity of $t \mapsto |s_2(t)|$, are not proved. Only existence (in $(-1, 0)$, and in $(-0.9999, -0.0001)$ for $t \leq 589$) is.

2 Setting: making the equation algebraic

The single most useful observation for the whole development is that the degree-6 polynomial of vf4 §5.1 is a perfect cube:

$$8s^6 - 24s^4 + 24s^2 - 8 \;=\; 8\,(s^2-1)^3. \tag{1}$$

Since $|8v^3|^{2/3} = 4v^2$ for real v (`abs_cube_rpow`), the awkward 2/3-power in E is a polynomial after all.

Theorem 2.1 (`E_eq`)**.** *For all $s, k_3 \in \mathbb{R}$,*

$$E(s,k_3) \;=\; 4\Big(s^4 + 24\,k_3^2\,(s^2-1)^2\,(s^6-3s^4-1)\Big),$$

and equivalently, with $u = s^2$ and $g(u) = 3u^2 - u^3 + 1$,

$$E(s,k_3) \;=\; 4\Big(s^4 - 24\,k_3^2\,(s^2-1)^2\,(3s^4 - s^6 + 1)\Big) \;=\; 4\big(u^2 - 24\,k_3^2\,(1-u)^2 g(u)\big).$$

Everything about signs of E is therefore rational arithmetic in s and k_3^2, and $E(\cdot, k_3)$ is continuous (`continuous_E`, proved by rewriting with Theorem 2.1 and `fun_prop`). In particular the paper's transcendental looking equation hides no transcendental behaviour: k_3 enters only through k_3^2.

3 Item 1: the root $s_2(t)$

3.1 Existence in $(-1, 0)$

Theorem 3.1 (`exists_root_unit`)**.** *For every $k_3 \neq 0$ there is $s \in (-1,0)$ with $E(s,k_3) = 0$.*

Proof. By Theorem 2.1, $E(-1,k_3) = 4 > 0$ (`E_neg_one`) and $E(0,k_3) = -96k_3^2 < 0$ (`E_zero`). Continuity on $[-1,0]$ and the intermediate value theorem (`intermediate_value_Ioo'`, the decreasing-orientation form) give a zero in the open interval. □

This is unconditional in t: the root the paper's `brentq` looks for exists for every $t \geq 1$, indeed for every real $k_3 \neq 0$.

3.2 The narrow interval of the paper, exactly

Write $u = s^2$, $u_0 = (0.9999)^2 = 99980001/10^8$, $u_1 = (10^{-4})^2 = 10^{-8}$, and $g(u) = 3u^2 - u^3 + 1 > 0$ on $(0,1)$. By Theorem 2.1,

$$E(s,k_3) > 0 \iff u^2 > 24\,k_3^2 (1-u)^2 g(u).$$

At the two endpoints of the paper's bracket:

- $s = -10^{-4}$: $u = 10^{-8}$, so the left side of the criterion is 10^{-16} while the right side is at least $24\,t^2\,(1-10^{-8})^2 \cdot 1$, which exceeds 10^{-16} for every $t \geq 1$. Hence $E(-10^{-4}, -t) < 0$ for all $t \geq 1$ (`E_right_neg`).
- $s = -0.9999$: the inequality is $t^2 < u_0^2/(24(1-u_0)^2 g(u_0))$, and the right-hand side equals $347\,187.6\ldots$, whose square root is $589.2261\ldots$. Since g and $u \mapsto u/(1-u)$ are increasing, the criterion is monotone in t^2, so $E(-0.9999, -t) > 0$ holds exactly for $t \leq 589$: `E_left_pos` for $t \leq 589$ and `E_left_neg` for $t \geq 590$.

Consequently:

Theorem 3.2 (`exists_root_grid`, `s2_spec`)**.** *For every integer t with $1 \leq t \leq 589$ there is $s \in (-0.9999, -0.0001)$ with $E(s,-t) = 0$; the root selector `s2` returns such an s and satisfies $E(s_2(t), -t) = 0$ and $|s_2(t)| < 1$.*

Remark 3.3 (The reported failure). The paper's prose asserts the root lies in $(-0.9999, -0.0001)$ for all $t \geq 1$ (§7.2, §6.5.2.1). As a statement about the *bracketing interval* this is false from $t = 590$ on: both endpoint values are then negative, so `brentq` on $[-0.9999, -0.0001]$ has no sign change to work with. This is `E_left_neg`; the endpoint values are

$$E(-0.9999, -589) = 0.0030691\ldots > 0, \qquad E(-0.9999, -590) = -0.0105088\ldots < 0.$$

The mechanism is transparent: the root satisfies $1 - u \approx (24\, t^2 g(u))^{-1/2}$, so it drifts to $s = -1$ like $1 - \frac{1}{4.9\, t}$ and leaves $(0.9999, 1)$ in modulus at $t \approx 589.23$. A bisection check over $t = 590, \ldots, 2000$ finds no zero of $E(\cdot, -t)$ in the interval at all, and the monotonicity of $u \mapsto u^2/((1-u)^2 g(u))$ on $(0, 1)$ (log-derivative $2/(u(1-u)) - (6u - 3u^2)/g > 0$) proves the same on paper; only the failure of the *sign change* is formalized.

None of this affects the paper's computations, which use $t \leq t_{\max} = 100$; the theorem above covers $1 \leq t \leq 589$ and hence the entire computational range with a factor 5.9 of margin.

Remark 3.4 (A numerical discrepancy in vf4 §7.3–7.4). The examples of vf4 §7.3 and §7.4 set `t = 1` but hard-code `s2 = -0.99941091795`. That value is the root at $t = 100$: $s_2(100) = -0.999410917960\ldots$, whereas $s_2(1) = -0.942823289665\ldots$. The hard-coded number is therefore inconsistent with the stated `t`; all quantitative statements in those two examples are unaffected only if one reads them at $t = 100$.

t	$E(-0.9999, -t)$	$E(-10^{-4}, -t)$	root $s_2(t)$ (bisection, 30 digits)
1	3.99839	-95.999998	$-0.942823289665459496 05$
10	3.99725	-9599.99981	$-0.994124806039169127 85$
100	3.88323	-959999.98	$-0.999410917960267812 77$
589	0.00307	$-3.3304 \cdot 10^7$	$-0.9998999616\ldots$
590	-0.01051	$-3.3418 \cdot 10^7$	none in $(-0.9999, -10^{-4})$

Table 1: The sign change of the paper's bracket. Left endpoint positive up to $t = 589$, negative from $t = 590$; right endpoint always negative.

4 Item 2: the lattice step and $N_{\text{phase}} = 0$

Theorem 4.1 (`b_eq_affine`, `b_step`, `b_step_uniform`). *For all $t \in \mathbb{N}$, $k \in \mathbb{Z}$ and $\sigma \in \mathbb{R}$,*

$$b(t, k, \sigma) = \frac{\arcsin(\sigma |s_2(t)|)}{\ln 2} - \frac{2\pi}{\ln 2}\, k, \qquad b(t, k+1, \sigma) - b(t, k, \sigma) = -\frac{2\pi}{\ln 2},$$

and the right-hand step is the same for all $t, t', k, k', \sigma, \sigma'$.

Theorem 4.2 (`b_strict_Anti`, `b_injective`). *For fixed t, σ the map $k \mapsto b(t, k, \sigma)$ is strictly antitone, hence injective.*

Proof. By the affine form the map is $k \mapsto c - \lambda k$ with $\lambda = 2\pi/\ln 2 > 0$. □

The phase defect is $\delta b_{t,k,\sigma} = b(t, k+1, \sigma) - b(t, k, \sigma) + 2\pi/\ln 2$; it is zero by the previous theorem, so the paper's N_{phase} vanishes identically:

Theorem 4.3 (`phase_Defect_eq_zero`, `Nphase_eq_zero`). *$\delta b_{t,k,\sigma} = 0$ for all t, k, σ, and for every finite set S of triplets the count $N_{\text{phase}}(S) = \#\{(t, k, \sigma) \in S : \delta b_{t,k,\sigma} \neq 0\}$ is 0.*

5 Item 3: counting the resonant stratum

5.1 Two elementary counting lemmas

Both are statements about finite sets of *integers* in a real interval, and both are proved from `Int.card_Icc` plus the floor/ceiling estimates $\lfloor b\rfloor \le b < \lfloor b\rfloor + 1$ and $\lceil a\rceil - 1 < a \le \lceil a\rceil$.

Lemma 5.1 (`card_le_of_forall_mem_Icc`). *If $S \subseteq \mathbb{Z}$ is finite and $S \subseteq [a, b]$, then $\#S \le b-a+1$.*

Lemma 5.2 (`le_card_of_forall_Icc_mem`). *If $S \subseteq \mathbb{Z}$ is finite and contains every integer of $[a, b]$, then $\#S \ge b - a - 2$.*

5.2 One lattice

For a phase α put $b_\alpha(k) = (\alpha - 2k\pi)/\ln 2$ and

$$k_{\mathrm{lo}} = \frac{\alpha - T\ln 2}{2\pi}, \qquad k_{\mathrm{hi}} = \frac{\alpha + T\ln 2}{2\pi}, \qquad k_{\mathrm{hi}} - k_{\mathrm{lo}} = \frac{T\ln 2}{\pi}.$$

The index set is $\texttt{kSet}(\alpha, T) = \{k \in \mathbb{Z} : \lceil k_{\mathrm{lo}}\rceil \le k \le \lfloor k_{\mathrm{hi}}\rfloor,\ k \ne 0\}$, and `mem_Icc_iff_abs_bOf` records the exact equivalence $k_{\mathrm{lo}} \le k \le k_{\mathrm{hi}} \iff |b_\alpha(k)| \le T$ (no side condition on T, only $\ln 2 > 0$, $\pi > 0$).

Theorem 5.3 (`card_kSet_bounds`). *For every phase α and $T \ge 0$,*

$$\Big|\#\{k \ne 0 :\ |b_\alpha(k)| \le T\} - \frac{T\ln 2}{\pi}\Big| \le 3.$$

Proof. Upper bound from Lemma 1 applied to $k_{Set} \subseteq [k_{lo}, k_{hi}]$; lower bound from Lemma 2 applied to $\{0\} \cup k_{Set}$, which contains all of $[k_{lo}, k_{hi}] \cap \mathbb{Z}$, and $\#(\{0\} \cup k_{Set}) \le \#k_{Set} + 1$. The constant 3 absorbs the floor, the ceiling and the excluded index $k = 0$. □

5.3 The 200 lattices and the slope

The family is $1 \le t \le 100$, $\sigma \in \{-1, 1\}$, i.e. 200 lattices (`card_latticeSet`). The triplet set $\texttt{modeSet}(T)$ and the two counts

$$N_{\mathrm{tot}}(T) = \#\mathrm{modeSet}(T), \qquad N_R(T) = \#\{b(t,k,\sigma) : (t,k,\sigma) \in \mathrm{modeSet}(T)\}$$

are defined exactly as in vf4 §6.4.2.

Theorem 5.4 (`Ntotal_bounds`). *For $T \ge 0$, $\big|N_{\mathrm{tot}}(T) - 200\,T\ln 2/\pi\big| \le 600$.*

Theorem 5.5 (`slope_bounds`, `Ntotal_approx`, `NR_le_slope`). $44.12 < 200\ln 2/\pi < 44.14$*; for $T \ge 0$,*

$$\big|N_{\mathrm{tot}}(T) - 44.13\,T\big| \le 600 + \tfrac{T}{100}, \qquad N_R(T) \le 44.14\,T + 600.$$

The numerical constants are the standard rational enclosures $3.141592 < \pi < 3.141593$ and $0.6931471803 < \ln 2 < 0.6931471808$.

5.4 Distinct count versus mode count

Theorem 5.6 (`NR_le_Ntotal`, `NR_eq_Ntotal_iff`). $N_R(T) \le N_{\mathrm{tot}}(T)$ *for every T, and*

$$N_R(T) = N_{\mathrm{tot}}(T) \iff \textit{no two distinct triplets of } \mathrm{modeSet}(T) \textit{ have the same } b.$$

Proof. modeSet(T) is finite; N_R is the cardinality of its image under $(t,k,\sigma) \mapsto b(t,k,\sigma)$, so this is `Finset.card_image_le` and `Finset.card_image_iff` (equality iff the map is injective on the set). □

The interesting point is that the lattice itself cannot collide. Formalizing this makes the equality criterion informative rather than vacuous.

Theorem 5.7 (`b_collision_structure`, `collision_imp_duplicate_phase`)**.** *Suppose* $1 \le t, t' \le 589$*,* $\sigma, \sigma' \in \{-1, 1\}$ *and* $b(t, k, \sigma) = b(t', k', \sigma')$*. Then*

$$\sigma = \sigma', \qquad |s_2(t)| = |s_2(t')|, \qquad k = k'.$$

In particular a collision in `modeSet`(T) *forces two distinct* $t \neq t'$ *with* $|s_2(t)| = |s_2(t')|$*.*

Proof. Multiplying the equality by $\ln 2$ and using the affine form, $\arcsin(\sigma|s_2 t|) - \arcsin(\sigma'|s_2 t'|) = 2\pi(k - k')$. Each arcsine lies in $[-\pi/2, \pi/2]$, so the left side has modulus at most π; hence $|k - k'| \le 1/2$, and as $k - k' \in \mathbb{Z}$ we get $k = k'$. Then the two arcsines are equal, and arcsin is injective on $[-1, 1]$, so $\sigma|s_2 t| = \sigma'|s_2 t'|$; as $|s_2| > 0$ (`s2_abs_pos`) and $\sigma, \sigma' = \pm 1$, this forces $\sigma = \sigma'$ and $|s_2 t| = |s_2 t'|$. □

Remark 5.8 (Reproducing the paper's table). Evaluating the definitions of this section with the true roots $s_2(t)$ reproduces vf4 §6.4.3 exactly, and with *zero* collisions, so that $N_R = N_{\text{tot}}$ there:

T	100	500	1000	5000	10000
$N_{\text{tot}} = N_R$	4200	21800	44000	220400	441000
N_R/T	42.0	43.6	44.0	44.08	44.10

The two smallest rows are below the asymptotic constant 44.127 because of the $O(1)$ correction of Theorem 5.3, not because of a modelling difference. The paper de-duplicates by `round(b,12)`; we use exact equality of real numbers. Since the lattice is collision-free by the rigidity theorem, the two agree here.

6 Item 4: the off-diagonal Gram bound

The Gram coefficient of vf4 §6.5.2.1 is defined as

$$G_{12}(T) = \begin{cases} 1, & b_1 = b_2, \\ \dfrac{e^{i(b_1 - b_2)T} - 1}{i(b_1 - b_2)T}, & b_1 \neq b_2. \end{cases}$$

Theorem 6.1 (`gramCoeff_offdiag_bound`)**.** *For* $b_1 \neq b_2$ *and* $T > 0$*,*

$$|G_{12}(T)| \;\le\; \frac{2}{T\,|b_1 - b_2|}.$$

Proof. $\|e^{i\theta} - 1\| \le \|e^{i\theta}\| + 1 = 2$ (`norm_exp_I_sub_one_le_two`, from $\|e^{i\theta}\| = 1$), and $\|i\theta\| = |\theta| = T|b_1 - b_2|$; divide. □

Remark 6.2. The diagonal case is $G_{11}(T) = 1$ (`gramCoeff_self`). The bound is the paper's Lemma 2 and is what makes the off-diagonal terms summable in $1/(T\delta_X)$; we do not formalize the summation, only the estimate, as requested.

7 Item 5: invariance of $\operatorname{Re} S_f$ under $b \mapsto -b$

For $N \ge 1$ and real b let

$$S_f(N, b) = \sum_{m=1}^{N} m^{-(1/2 + ib)},$$

formalized as `hsum` with the complex power `Complex.cpow` on the natural cast $m \mapsto (m : \mathbb{C})$; the summand is `hterm m b`.

Theorem 7.1 (`hterm_conj`, `hsum_conj`, `hsum_re_even`)**.** *For all $N \geq 1$ and $b \in \mathbb{R}$,*

$$\overline{S_f(N,b)} = S_f(N,-b), \qquad \textit{hence} \qquad \operatorname{Re} S_f(N,-b) = \operatorname{Re} S_f(N,b).$$

Proof. $\overline{m^z} = m^{\bar{z}}$ for the positive real m, because conjugation is continuous and fixes m (`Complex.cpow_conj`, applicable since $\arg(m : \mathbb{C}) = 0 \neq \pi$); $\overline{1/2 + ib} = 1/2 - ib$; sum over $m \leq N$ using that conjugation is an additive map. □

The consequence is the one the paper uses when it treats $\sigma = \pm 1$ symmetrically: the sign parameter only reflects the grid, and the real part of the partial sums cannot distinguish b from $-b$.

8 Item 6: the $6n \pm 1$ filter

Theorem 8.1 (`prime_mod_six`, `prime_zmod_six`)**.** *If p is prime and $p > 3$, then $p \equiv 1$ or $p \equiv 5 \pmod 6$; equivalently the class of p in $\mathbb{Z}/6$ is 1 or 5.*

Proof. p is odd, so $2 \nmid p$, and $3 \nmid p$; hence $p \bmod 6 \notin \{0, 2, 3, 4\}$ (each of those residues is divisible by 2 or by 3), leaving 1 or 5 among $0, \ldots, 5$. In Lean: `Nat.Prime.eq_one_or_self_of_dvd` converts $2 \mid p$ and $3 \mid p$ into $p = 2$ resp. $p = 3$, contradicting $p > 3$; `omega` then finishes. □

Theorem 8.2 (`converse_fails`, `filter_not_sufficient`)**.** $25 \equiv 1 \pmod 6$ *and 25 is not prime; thus the filter alone is not a primality test.*

9 The Gram matrix, N_{off} and N_{van}

vf4 §6.4.4–6.4.8 defines the two spectral diagnostics numerically, from the Gram matrix of the resonant modes. `FrequencyGrids/Eigen.lean` formalizes exactly those objects.

The matrix

The modes are the triplets of `modeSet`(T) (§5); since collisions are rigid (Theorem 5.7), distinct triplets carry distinct frequencies, so indexing by triplets and by distinct frequencies agree. Write b_i for the frequency of the mode i (`modeB`).

Definition 9.1 (`gramMatrix`)**.**

$$G_{ij}(X) = \sum_{n \leq X} n^{-1-i(b_i - b_j)}, \qquad i, j \in \operatorname{modeSet}(T),$$

as a matrix over $\mathbb{C}$ indexed by the (finite) triplet set; the sum runs over $1 \leq n \leq \lfloor X \rfloor$.

Theorem 9.2 (`gramMatrix_isHermitian`)**.** $G(X)^{\mathsf{H}} = G(X)$. *Hence the symmetrisation $(G + G^{\mathsf{H}})/2$ of the code is the identity on G.*

Proof. $\overline{G_{ji}} = \sum_n \overline{n^{-1-i(b_j - b_i)}} = \sum_n n^{-1+i(b_j - b_i)} = \sum_n n^{-1-i(b_i - b_j)} = G_{ij}$, using $\overline{m^w} = m^{\bar{w}}$ for $m \in \mathbb{N}$ (`cpow_conj_natCast`, from `Complex.cpow_conj` with $\arg(m) = 0 \neq \pi$) and $\overline{-1 - i(x-y)} = -1 - i(y-x)$. □

The spectrum

Definition 9.3 (`eig`, `eigPos`, `eigMax`)**.** λ_j are the eigenvalues of the Hermitian matrix $G(X)$ given by the spectral theorem (`Matrix.IsHermitian.eigenvalues`), hence real; $\lambda_j^+ = \max(\lambda_j, 0)$ is the clamped eigenvalue, exactly `np.maximum(eigenvalues, 0)` of vf4 §6.4.7; and $\max_j \lambda_j^+$ is the supremum of the range, with $\sup \emptyset = 0$ matching the code's empty-matrix branch.

Remark 9.4. The clamp is the code's protective maximum, and it is now *proved* to be the identity: Theorem 9.11 below factors $G = EE^{\mathsf{H}}$, so the eigenvalues are nonnegative and $\max(\lambda_j, 0) = \lambda_j$ (`eigPos_eq_eig`). The definition keeps the maximum, but it never fires.

The two counts

Definition 9.5 (`offSum`, `offRatio`, `Noff`)**.** $I_j(X) = \sum_{i\neq j} |G_{ij}(X)|$, $r_j(X) = I_j(X)/|G_{jj}(X)|$, and

$$N_{\mathrm{off}}(T, X, \delta_X) = \#\{j : \ r_j(X) > \delta_X\}.$$

(As G is Hermitian, $|G_{ij}| = |G_{ji}|$, so the orientation of the sum is immaterial.)

Definition 9.6 (`Nvan`)**.**

$$N_{\mathrm{van}}(T, X, \delta_T) = \#\{j : \ \lambda_j^+ \leq \delta_T \cdot \max_i \lambda_i^+\}.$$

Theorem 9.7 (`Noff_le_Ntotal`, `Nvan_le_Ntotal`)**.** *Both counts are at most the number of modes:* $N_{\mathrm{off}}(T, X, \delta) \leq N_{\mathrm{tot}}(T)$ *and* $N_{\mathrm{van}}(T, X, \delta) \leq N_{\mathrm{tot}}(T)$*. Moreover* $\lambda_j^+ \leq \max_i \lambda_i^+$ *for every* j*, and the threshold is nonnegative when there is at least one mode (`eigPos_le_eigMax`, `eigMax_nonneg`).*

Theorem 9.8 (`etaR_eq`)**.** *With these definitions and the proved* $N_{\mathrm{phase}} = 0$*,*

$$\eta_R(T, X, \delta) = 1 - \frac{N_{\mathrm{off}}(T, X, \delta) + N_{\mathrm{van}}(T, X, \delta)}{N_R(T)}.$$

9.1 The factorisation, the spectrum and the rank

Definition 9.9 (`gramE`)**.** $E : \ \mathrm{modeSet}(T) \times \{0, \ldots, \lfloor X \rfloor - 1\} \to \mathbb{C}, \quad E_{i,m} = (m+1)^{-1/2-i\,b_i}$.

Theorem 9.10 (`gramMatrix_eq_mul_conjTranspose`)**.** $G(X) = E\,E^{\mathsf{H}}$*; that is,* $G_{ij}(X) = \sum_{m<\lfloor X \rfloor} E_{i,m}\overline{E_{j,m}}$.

Proof. $E_{i,m}\overline{E_{j,m}} = n^{-1/2-ib_i} n^{-1/2+ib_j} = n^{-1-i(b_i-b_j)}$ with $n = m+1$, using $\overline{n^w} = n^{\bar{w}}$ and $n^a n^b = n^{a+b}$ for $n \neq 0$. □

Theorem 9.11 (`gramMatrix_posSemidef`, `eig_nonneg`, `eigPos_eq_eig`, `rank_gramMatrix_le`)**.** $G(X)$ *is positive semidefinite; every eigenvalue is* ≥ 0*; the clamp is the identity; and* $\operatorname{rank} G(X) \leq \lfloor X \rfloor$.

Proof. $EE^{\mathsf{H}} \succeq 0$ (`Matrix.posSemidef_self_mul_conjTranspose`), and $\operatorname{rank}(EE^{\mathsf{H}}) \leq \operatorname{rank} E \leq \lfloor X \rfloor$ since E has $\lfloor X \rfloor$ columns. Eigenvalues are those of the Hermitian matrix, so the nonnegativity is `PosSemidef.eigenvalues_nonneg` — transported along proof irrelevance of the hermitianity witness, since the eigenvalue map has a proof-independent value type. □

Theorem 9.12 (`card_nonzero_eig_eq_rank`)**.** *The number of nonzero eigenvalues is the rank:* $\#\{j : \lambda_j \neq 0\} = \operatorname{rank} G(X)$*. Equivalently, the zero eigenvalues number exactly* $\#\mathrm{modeSet}(T) - \operatorname{rank} G(X)$.

Theorem 9.13 (`card_le_Nvan_add`, `Ntotal_sub_le_Nvan`, `NR_sub_le_Nvan`)**.** *For every* $\delta \geq 0$*,*

$$\#\mathrm{modeSet}(T) \ \leq \ N_{\mathrm{van}}(T, X, \delta) + \lfloor X \rfloor, \qquad \textit{i.e.} \qquad N_{\mathrm{van}}(T, X, \delta) \ \geq \ N_{\mathrm{tot}}(T) - \lfloor X \rfloor.$$

The same holds with N_R *in place of* N_{tot}*, because* $N_R = N_{\mathrm{tot}}$ *(Theorem 9.18).*

Proof. Every zero eigenvalue satisfies $0 \leq \delta \max_j \lambda_j^+$ (the threshold is nonnegative), so the vanishing count contains the zero-eigenvalue count; the nonzero eigenvalues are exactly $\operatorname{rank} G(X) \leq \lfloor X \rfloor$ many. □

9.2 The general frequency family, and the distinct-frequency lattice

Nothing in the factorisation uses that the frequencies come from triplets: it only uses that column m of E is $(m+1)^{-1/2-if_i}$. So the whole spectral package is proved once for an arbitrary finite family.

Definition 9.14 (`gramOf`, `evalOf`, `evalFin`)**.** For $f : \iota \to \mathbb{R}$ with ι finite,

$$G_{ij}(X) = \sum_{n \le X} n^{-1-i(f_i - f_j)}, \qquad E_{i,m} = (m+1)^{-1/2-i\,f_i},$$

the latter with index $m : \mathrm{Fin}\,\lfloor X \rfloor$ (`evalOf`) or $m : \mathrm{Fin}\,X$ for a natural cutoff (`evalFin`).

Theorem 9.15 (`gramOf_eq_mul_conjTranspose`, `gramOf_isHermitian`, `gramOf_posSemidef`, `gramOf_diag`)**.** *$G = EE^{\mathsf{H}}$ (and, with the natural cutoff, `gramOf_nat_eq_mul_conjTranspose` gives the same with E indexed by $\mathrm{Fin}\,X$); G is Hermitian and positive semidefinite; and the diagonal is the harmonic number, $G_{ii} = H_{\lfloor X \rfloor}$.*

Theorem 9.16 (`eigOf_nonneg`, `rank_gramOf_le_rank_eval`, `rank_gramOf_le`, `card_nonzero_eigOf_eq_rank`, `vanishOf_ge`)**.** *All eigenvalues are ≥ 0; $\mathrm{rank}\,G \le \mathrm{rank}\,E \le \lfloor X \rfloor$; the nonzero eigenvalues number exactly the rank; hence for every $\delta \ge 0$*

$$\#\iota \;\le\; N_{\mathrm{van}}(f, X, \delta) + \lfloor X \rfloor.$$

Two instances. First, '`gramMatrix_eq_gramOf`' records that the triplet-indexed matrix of §9.1 *is* the instance $f = \mathtt{modeB}\,T$ (definitionally). Second, and this is the point of the generalisation, take

$$\iota = \mathtt{freqSet}(T) = \{b(t,k,\sigma)\} \text{ (distinct values)}, \qquad \#\iota = N_R(T),$$

which is the indexation the paper's code actually uses after de-duplication. Collision rigidity (Theorem 5.7) says that distinct triplets have distinct frequencies unless $t \mapsto |s_2(t)|$ repeats, so the two indexings differ only in that degenerate case.

Theorem 9.17 (`card_freqSet`, `gramFreq_isHermitian`, `gramFreq_posSemidef`, `rank_gramFreq_le`, `NvanFreq_ge`, `NvanFreq_sub_ge`)**.** *Let $G_{\mathrm{freq}}(T, X)$ be the Gram matrix indexed by `freqSet`(T). Then $\#\iota = N_R(T)$, G_{freq} is Hermitian and positive semidefinite,* $\mathrm{rank}\,G_{\mathrm{freq}} \le \lfloor X \rfloor$*, and for every $\delta \ge 0$*

$$N_{\mathrm{van}}(F, T, X, \delta) \;\ge\; N_R(T) - \lfloor X \rfloor$$

with no hypothesis on collisions. *This strictly improves Theorem 9.13, whose N_R-form had to assume that no collision occurs.*

9.3 Collisions are impossible

Rigidity (`b_collision_structure`) reduces a collision to $\sigma = \sigma'$, $k = k'$ and $|s_2(t)| = |s_2(t')|$. The last equality cannot hold for $t \ne t'$: both roots are negative, so it makes them one real number s, and then $E(s, -t) = 0 = E(s, -t')$ reads

$$u^2 \;=\; 24\,t^2(1-u)^2(1+3u^2-u^3) \;=\; 24\,t'^2(1-u)^2(1+3u^2-u^3), \qquad u = s^2 \in (0,1),$$

and $(1-u)^2(1+3u^2-u^3) > 0$, so $t^2 = t'^2$.

Theorem 9.18 (`abs_s2_injective`, `not_hasCollision`, `NR_eq_Ntotal`)**.** *For $1 \le t, t' \le 589$, $|s_2(t)| = |s_2(t')|$ implies $t = t'$. Hence $\neg\mathrm{HasCollision}(T)$ for every T, and $N_R(T) = N_{\mathrm{tot}}(T)$ unconditionally.*

This removes every "no collision" hypothesis from the earlier statements (`NR_sub_le_Nvan`, `NvanLittleO.eventually_le_NR`, `not_NvanLittleO_of_le_NR`). Note what is *not* claimed: uniqueness of the root $s_2(t)$ itself. The argument above only needs that two parameters cannot share one root, which is weaker than uniqueness for each parameter, and needs no monotonicity and no calculus.

9.4 A numerical remark: the diagnostics are not small

The point of replacing opaque symbols by the real definitions is that the numbers can now be computed. Evaluating them with the paper's own parameters for the yield experiment (vf4 §6.4.8: $t_{\max} = 10$, $X = 5000$, $\delta = 10^{-2}$) gives:

T	N_R	N_{off}	N_{off}/N_R	N_{van}/N_R
50	200	200	1.000	0.865
100	420	420	1.000	0.890
200	860	860	1.000	0.898
500	2180	2180	1.000	0.900

At $T = 100$ the ratio N_{van}/N_R is stable under X: 0.898, 0.898, 0.890, 0.883 for $X = 200$, 10^3, $5 \cdot 10^3$, $2 \cdot 10^4$. In other words the sequences $a_T = N_{\text{phase}}/N_R$, $b_T = N_{\text{off}}/N_R$, $c_T = N_{\text{van}}/N_R$ of vf4 §6.4.8 read 0, 1.000 and ≈ 0.9 here, not $0, 0, 0$; the conjecture $N_{\text{off}}, N_{\text{van}} = o(N_R)$ is not what these definitions produce at the paper's parameters.

Two mechanisms, both elementary, explain the numbers. Neither is formalized; both are checked numerically.

- *Why $N_{\text{off}} = N_R$.* The normalisation is the harmonic sum $|G_{jj}| = H_X = \sum_{n \le X} n^{-1} \approx \ln X$ (here 9.09) — this is Theorem 9.2 of §9.1 again: the diagonal entry is exactly $H_{\lfloor X \rfloor}$ (`gramMatrix_diag`, and Mathlib's `harmonic` in `gramMatrix_diag_eq_harmonic`) — while the numerator $I_j = \sum_{i \ne j} |G_{ij}|$ is a sum of *absolute values* — there is no cancellation to exploit. For a neighbour at frequency distance δ_b with $\delta_b \ln X \ll 1$ one has $|G_{ij}| \approx \log(1/\delta_b) + O(1) \approx H_X$; with density $\approx 2t_{\max} \ln 2/\pi \approx 4.4$ modes per unit frequency the typical gap is 0.017 and the typical nearest-neighbour entry is $|G_{ij}| \approx 8.5$. Measured at $T = 100$, $X = 5000$: $\min_j r_j = 64.3$, so every mode exceeds a fixed tolerance $\delta < 1$.
- *Why N_{van} is large.* Since $G = AA^*$, $\operatorname{rank} G \le X$, so at least $N_R - X$ eigenvalues vanish identically: $N_{\text{van}} \ge N_R - X$ whenever $\delta_T \ge 0$. Measured at $T = 100$, $X = 300$ ($N_R = 4200$): $N_{\text{van}} = 4158 \ge 3900 = N_R - X$. When $X > N_R$ no such bound applies, but A is a modulated Vandermonde matrix in the points $\log_2 n$ whose spectrum decays quickly, so most eigenvalues still sit below $\delta_T \max_j \lambda_j$; at $T = 100$, $X = 5000$, 336 of the 420 clamped eigenvalues (80%) are below $10^{-3} \max_j \lambda_j$, and 374 (89%) below the paper's $\delta_T = 10^{-2}$.

The statement $N_{\text{off}} = \#\text{modeSet}$ is not proved — but its analytic input now is. At the parameters of vf4 §6.4.8 the numbers above say $r_j(X) > 10^{-2}$ for every mode. Section 12 proves, with no asymptotics, that a *single* neighbour within $1.55/\log X$ already forces $r_j \ge 1/50$, hence $N_{\text{off}} = \#\text{modeSet}$ for every $\delta < 1/50$ (`Noff_eq_card_of_close`). What is missing is the finite geometric verification: that every mode at $t_{\max} = 10$, $X = 5000$ has such a neighbour (within ≈ 0.182 in frequency). That needs two things this development does not yet have: certified rational enclosures of $s_2(1), \ldots, s_2(10)$ (see Remark 12.5) and a parameterized $t_{\max}$.

None of this proves the conjectures false: they quantify over $T \to \infty$ together with $X(T) \to \infty$ and $\delta_T \to 0$, and a differently scaled tolerance could change the counts. But it does mean the numerical evidence adduced in vf4 §6.4.8–6.4.10 is not evidence for them, and that any repair must change the definitions (for N_{off} the tolerance would have to grow with N_R; for N_{van} the threshold must be tied to the conditioning or X must dominate N_R).

10 The truncation cannot grow slowly

Theorem 9.13 is the whole content of the next result: since a mode with zero Gram eigenvalue is always counted by N_{van}, and there are at least $\#\text{modeSet}(T) - X$ of those, the vanishing count can only be negligible if the truncation is not.

Theorem 10.1 (`NvanLittleO.eventually_le_Ntotal`)**.** *If $N_{\mathrm{van}}(T, X(T), \delta(T)) = o(N_R(T))$, then for every $\varepsilon > 0$ and all sufficiently large T,*

$$X(T) \;\geq\; (1-\varepsilon)\, N_{\mathrm{tot}}(T).$$

Proof. Fix $\varepsilon > 0$ (for $\varepsilon \geq 1$ there is nothing to prove). Take $\delta \equiv 0$ in the little-o hypothesis and $c = \varepsilon/2$: eventually $N_{\mathrm{van}}(T, X(T), 0) \leq c\, N_R(T)$. By Theorem 9.13,

$$N_{\mathrm{tot}}(T) \;\leq\; N_{\mathrm{van}}(T, X(T), 0) + X(T) \;\leq\; c\, N_R(T) + X(T) \;\leq\; c\, N_{\mathrm{tot}}(T) + X(T),$$

using $N_R \leq N_{\mathrm{tot}}$. Hence $(1-c)N_{\mathrm{tot}}(T) \leq X(T)$, and $1 - \varepsilon \leq 1 - c$. □

Theorem 10.2 (`NvanLittleO.eventually_le_NR`)**.** *Under the same hypothesis, $X(T) \geq (1-\varepsilon)N_R(T)$ for all large T, for every $\varepsilon > 0$ — with no hypothesis on collisions, since there are none (Theorem 9.18).*

The contrapositive is the quantitative statement the paper's conjecture is committed to:

Theorem 10.3 (`not_NvanLittleO_of_le_Ntotal`, `not_NvanLittleO_of_le_NR`)**.** *Let $c < 1$. If $X(T) \leq c\, N_{\mathrm{tot}}(T)$ for all sufficiently large T, then $N_{\mathrm{van}} \neq o(N_R)$. Since $N_R = N_{\mathrm{tot}}$ (`NR_eq_Ntotal`), the same conclusion holds with N_R in place of N_{tot}.*

Proof. Otherwise Theorem 10.1 with $\varepsilon = (1-c)/2$ gives $(1-\varepsilon)N \leq X \leq c\, N$ with $1 - \varepsilon > c$ and $N > 0$ eventually (`eventually_Ntotal_pos`), a contradiction. □

For the distinct-frequency count of §9.2 the same conclusion holds with no hypothesis at all, because its lower bound is unconditional:

Theorem 10.4 (`NvanFreqLittleO.eventually_le_NR`, `not_NvanFreqLittleO_of_le_NR`)**.** *If $N_{\mathrm{van}}(F, T, X(T), \delta(T)) = o(N_R(T))$ then, for every $\varepsilon > 0$, $X(T) \geq (1-\varepsilon)N_R(T)$ for all large T. Consequently, if $c < 1$ and $X(T) \leq c\, N_R(T)$ eventually, then $N_{\mathrm{van}}(F) \neq o(N_R)$. Neither statement assumes anything about collisions (they use `NvanFreq_ge` and `eventually_NR_pos`).*

Since $N_{\mathrm{tot}}(T) \sim 200\, T \ln 2/\pi \approx 44.13\, T$ (§5), the conjecture of vf4 §6.4.13 therefore requires a truncation growing at least linearly in T, with constant at least 44.13: no fixed cutoff X can work, and neither can any $X = o(T)$. This is consistent with, and explains, the numbers of §9.4: at $X = 5000$ and $T = 500$ one has $N_R = 2180$, and indeed $N_{\mathrm{van}}/N_R = 0.9$.

11 The conjectures: stated, not proved

We reproduce the paper's conjectural layer verbatim as `def ... : Prop`, so that the statements are machine-checked formulations while remaining unproved. The two diagnostics occurring in them are the concrete definitions of §9.

Definition 11.1 (`etaR`)**.**

$$\eta_R(T, X, \delta) = 1 - \frac{N_{\mathrm{phase}} + N_{\mathrm{off}}(T, X, \delta) + N_{\mathrm{van}}(T, X, \delta)}{N_R(T)},$$

with $N_{\mathrm{off}}, N_{\mathrm{van}}$ as in §9 and $N_{\mathrm{phase}} = 0$ by Theorem 4.3, so that $\eta_R = 1 - (N_{\mathrm{off}} + N_{\mathrm{van}})/N_R$ (`etaR_eq`).

Conjecture (stated, not proved) 11.2 (`Eta_Tends_To_One`)**.** For every $X, \delta : \mathbb{R} \to \mathbb{R}$ with $X(T) \to \infty$ and $\delta(T) \to 0$, $\eta_R(T, X(T), \delta(T)) \to 1$ as $T \to \infty$.

Conjecture (stated, not proved) 11.3 (`Noff_LittleO`)**.** Along the same double limit, $N_{\mathrm{off}}(T, X(T), \delta(T)) = o(N_R(T))$.

Conjecture (stated, not proved) 11.4 (`Nvan_LittleO`)**.** Along the same double limit, $N_{\mathrm{van}}(T, X(T), \delta(T)) = o(N_R(T))$.

Conjecture (stated, not proved) 11.5 (`Primality_Criterion_Conjecture`)**.** For odd $n > 3$, with

$$S_f(N) = \sum_{\sigma\in\{-1,1\}} \sum_{0<|k|\le 10} \sum_{m\le N} m^{-(1/2+\,i\,b(1,k,\sigma))},$$

one has $\mathrm{Re}\big(S_f(n-1)\,S_f(n+1)\big) < 0$ if and only if n is prime.

Remark 11.6. Nothing in the development proves, assumes or uses these four propositions. In particular no theorem is stated conditionally on them, and `#print axioms` on the corresponding `def`s reports only `propext`, `Classical.choice`, `Quot.sound` — the definitions are inert.

12 An explicit estimate for the Gram entries

The numerical remark of §9.4 explains the size of N_{off} but proves nothing about it. The following estimate is elementary and unconditional — no asymptotics, no Euler–Maclaurin.

Theorem 12.1 (`cos_155_ge`)**.** $\cos 1.55 \ge 1/50$.

Proof. $1.55 < \pi/2$ and $\cos 1.55 = \sin(\pi/2 - 1.55)$; with $x = \pi/2 - 1.55 \in (0.0207, 0.025)$ the Taylor bound `Real.sin_gt_sub_cube` gives $\sin x > x - x^3/6 > 0.0207 - 3\cdot 10^{-6} > 1/50$, using $3.1415 < \pi < 3.15$. □

Theorem 12.2 (`re_cpow_gram`, `re_gramOf`)**.** $\mathrm{Re}\big(n^{-1-i\theta}\big) = n^{-1}\cos(\theta\log n)$, *hence*

$$\mathrm{Re}\,G_{ij}(X) = \sum_{n\le X\le} \frac{\cos((f_i - f_j)\log n)}{n}.$$

Theorem 12.3 (`re_gramOf_ge`, `norm_gramOf_ge`)**.** *If* $|f_i - f_j|\cdot\log X \le 1.55$ *and* $X \ge 1$*, then*

$$\mathrm{Re}\,G_{ij}(X) \;\ge\; \tfrac{1}{50}\,H_{\lfloor X\rfloor}, \qquad \textit{and hence} \qquad \|G_{ij}(X)\| \;\ge\; \tfrac{1}{50}\,H_{\lfloor X\rfloor}.$$

Proof. Termwise: $|\theta\log n| \le |\theta|\log X \le 1.55 < \pi/2$ for $n \le \lfloor X\rfloor$, and cos decreases on $[0, \pi/2]$, so $\cos(\theta\log n) \ge \cos 1.55 \ge 1/50$; multiply by $1/n > 0$ and sum. For the norm, $|\mathrm{Re}\,z| \le \|z\|$. □

Theorem 12.4 (`offRatio_ge_of_close`, `Noff_eq_card_of_close`)**.** *If every mode* j *has another mode* $i \ne j$ *with* $|b_i - b_j|\cdot\log X \le 1.55$*, then* $r_j(X) \ge 1/50$ *for every* j*, and consequently*

$$N_{\mathrm{off}}(T, X, \delta) \;=\; \#\mathrm{modeSet}(T) \qquad \textit{for every } \delta < 1/50,$$

in particular for $\delta = 10^{-2}$.

Proof. $r_j = I_j/H_{\lfloor X\rfloor} \ge |G_{ij}|/H_{\lfloor X\rfloor} \ge 1/50$ by the previous theorem and the harmonic diagonal; then every mode passes the threshold. □

Remark 12.5. At $X = 5000$ the geometric hypothesis reads $|b_i - b_j| \le 1.55/\log X \approx 0.182$ (`close_iff_div`). What is *not* proved is that this hypothesis holds at the parameters of vf4 §6.4.8: that is a finite verification over the 420 modes at $T = 100$ (and 2180 at $T = 500$), which needs certified rational enclosures of $s_2(1), \dots, s_2(10)$ and a parameterized $t_{\max}$. The reduction is formalized; the finite computation is not. See `paper/phase-status.md`.

13 Uniqueness of the root, and $t_{\max}$ as a parameter

13.1 Strict monotonicity and uniqueness

Write $u = s^2$. By `E_eq'`, $E(s,-t) = 0$ on $(-1,0)$ is equivalent to $f(u) = 24t^2$ for

$$f(u) = \frac{u^2}{(1-u)^2(1+3u^2-u^3)}, \qquad u \in (0,1).$$

Theorem 13.1 (`deriv_rootRat`, `logDeriv_ge_five`, `rootRat_strictMonoOn`)**.** *On $(0,1)$ (the identity for f'/f is `deriv_rootRat_log`),*

$$f'(u) = \frac{u(1-u)(2+7u^3-3u^4)}{\big((1-u)^2(1+3u^2-u^3)\big)^2} \quad \text{and} \quad \frac{f'}{f}(u) = \frac{2}{u(1-u)} - \frac{6u-3u^2}{1+3u^2-u^3} \;\geq\; 8-3 \;=\; 5 \;>\; 0.$$

Hence f is strictly increasing on $(0,1)$.

Proof. The two crude bounds are $2/(u(1-u)) \geq 8$ (since $u(1-u) \leq 1/4$) and $(6u-3u^2)/(1+3u^2-u^3) \leq 3$ (numerator ≤ 3, denominator ≥ 1). Strict monotonicity follows from `strictMonoOn_of_deriv_pos`. □

Theorem 13.2 (`root_iff`, `exists_unique_root_grid`, `s2_unique`, `abs_s2_strictMono`)**.** *For $1 \leq t \leq 589$ the equation $E(\cdot,-t) = 0$ has exactly one root in the paper's bracket $(-0.9999, -0.0001)$; any such root equals $s_2(t)$, so the definition of s_2 involves no genuine choice; and $t \mapsto |s_2(t)|$ is strictly increasing.*

Theorem 13.3 (`s2_mem_of_sign_change`)**.** *If $lo < hi$ in the bracket and $E(lo,-t) > 0 > E(hi,-t)$, then $s_2(t) \in (lo, hi)$. This is how a rational bracket, checked by `norm_num` on E, becomes a certified enclosure of $s_2(t)$.*

Theorem 13.4 (`s2_enclosure_1, ..., s2_enclosure_10`)**.** *$s_2(t)$ is bracketed by two rationals at distance $4 \cdot 10^{-6}$, for $t = 1, \ldots, 10$:*

$$s_2(1) \in (-0.942826, -0.942822), \quad \ldots, \quad s_2(10) \in (-0.994127, -0.994123).$$

Each bracket is checked by `norm_num` on the polynomial E at the two ends, and transported to $s_2(t)$ by `s2_mem_of_sign_change`.

13.2 The truncation $t_{\max}$

Theorem 13.5 (`latticeSetOf`, `card_latticeSetOf`, `modeSetOf`, `mem_modeSetOf`, `modeSet_eq`, `NtotalOf`, `NtotalOf_bounds`)**.** *For a parameter $t_{\max}$, the lattice $[1, t_{\max}] \times \{-1, 1\}$ has $2t_{\max}$ elements, the triplet set is characterised by $t \leq t_{\max}$, $\sigma = \pm 1$, $k \neq 0$, $|b| \leq T$, and*

$$\left| N_{\mathrm{tot}}(t_{\max}, T) - 2\, t_{\max}\, \frac{T \ln 2}{\pi} \right| \;\leq\; 6\, t_{\max}.$$

$t_{\max} = 100$ is the instance used throughout (`modeSet_eq`, `rfl`), so the paper's two experiments ($t_{\max} = 100$ in §6.4.3, $t_{\max} = 10$ in §6.4.8) are the two ends of one statement.

14 Verification notes

Axioms. For every theorem of items 1–6, `#print axioms` returns a subset of {`propext`, `Classical.choice`, `Quot.sound`}. In particular the obvious question — does the development lean on `N_off`, on choice principles beyond the classical ones, or on `sorry` — has a clean answer: no. There is no `sorry` and no `axiom` declaration in the sources.

Build. The files are compiled with the pre-built Mathlib of the environment:

```
export LEAN_PATH=$PWD:$LEAN_PATH
lean -o FrequencyGrids/Basic.olean -i FrequencyGrids/Basic.ilean \
     FrequencyGrids/Basic.lean
...                     # one line per module, see scripts/build.sh
./scripts/build.sh
```

The declaration table of §15 is generated by `paper/tools/build_paper.py` from a single list of rows, so that it can be kept in step with the sources.

`scripts/build.sh` compiles the modules in dependency order (`Basic` → `Lattice` → `Counting` → `Structure` → `Resonance` → `Congruence` → `Conjectures` → root). There is no Lake project: the development is a set of modules over Mathlib's own build.

Design choices worth knowing.

- `s2` is a *selector*, defined by classical choice from the existence theorem, and equals 0 outside $1 \le t \le 589$. Every statement that uses its properties carries the hypotheses $1 \le t \le 589$ explicitly; no statement silently assumes them. (The paper's computational range is $t \le 100$.)
- The triplet type is $\mathbb{N} \times \mathbb{Z} \times \mathbb{R}$: t is a natural, k an integer, σ a real constrained by $\sigma \in \{-1, 1\}$ (`sigmaSet`). Integers for k matter: injectivity of $k \mapsto b$ and the $\lceil\cdot\rceil/\lfloor\cdot\rfloor$ counting live on $\mathbb{Z}$.
- `latticeSet` hard-codes $t_{\max} = 100$; the paper's $t_{\max}$ is not a parameter. Making it a variable is a mechanical change (the constants 200 and 600 would become $2t_{\max}$ and $6t_{\max}$).
- The two counting lemmas are stated for arbitrary finite $S \subseteq \mathbb{Z}$ rather than for `kSet`, which keeps the floor/ceiling arithmetic out of the application.
- The positive-semidefiniteness of G is stated in Mathlib's sense, which needs an order on $\mathbb{C}$: the file opens the `ComplexOrder` scope for that, giving $\mathbb{C}$ its star-order ("$z \le w$ iff $w - z$ is a nonnegative real"). This is a scoped instance, so the surrounding development is unaffected.
- The spectral diagnostics are indexed by the triplet set modeSet(T) itself rather than by Fin N_R: this avoids building an enumeration, and the permutation-invariance of the spectrum makes the choice immaterial. Eigenvalues come from `Matrix.IsHermitian.eigenvalues`, i.e. from the spectral theorem, so they are exact reals — a diagonalisation cannot be computed in Lean, and nothing in the definitions needs it to be.

What a reader should not conclude. That vf4's spectral programme is validated. The formalization certifies the bookkeeping around the construction — the lattice step, the counting, the Gram estimate, the symmetry, the arithmetic filter, and the existence of the root — and proves two of the paper's implicit numerical claims false as stated (Remark 3.3, Remark 3.4). It proves nothing about $\eta_R(T) \to 1$ or about primality detection. What it does now settle is the *vocabulary*: the conjectures are stated about the paper's actual diagnostics (§9), and those diagnostics, evaluated at the paper's own parameters, are far from negligible (§9.4). The conjectures are therefore open statements whose published evidence does not survive reimplementation — which is a different, and weaker, position than the paper claims, not a refutation.

15 Declaration-by-declaration table

Every declaration of the development is listed below with its kind, its statement and its status. "proved" means a closed Lean proof with no `sorry`; "definition" means a definition; "not proved" means a `def ... : Prop` that intentionally has no proof.

Declaration	Kind	Statement	Status
FrequencyGrids/Basic.lean — the characteristic equation and the root $s_2(t)$ (item 1)			
char_Poly	def	$\mathrm{charPoly}(s) = 8s^6 - 24s^4 + 24s^2 - 8$	definition
char_Poly_eq	thm	$\mathrm{charPoly}(s) = 8(s^2-1)^3$	proved
abs_cube_rpow	thm	$\lvert 8v^3\rvert^{2/3} = 4v^2$ for $v \in \mathbb{R}$	proved
E	def	$E = \lvert\mathrm{charPoly}\rvert^{2/3}\cdot P + 4s^4$, with $P = 24k_3^2 s^6 - 72k_3^2 s^4 - 24k_3^2$	definition
E_eq	thm	$E(s,k_3) = 4\big(s^4 + 24k_3^2(s^2-1)^2(s^6-3s^4-1)\big)$	proved
E_eq'	thm	$E(s,k_3) = 4\big(s^4 - 24k_3^2(s^2-1)^2(3s^4-s^6+1)\big)$	proved
continuous_E	thm	$s \mapsto E(s,k_3)$ is continuous	proved
E_neg_one	thm	$E(-1,k_3) = 4$	proved
E_zero	thm	$E(0,k_3) = -96k_3^2$	proved
exists_root_unit	thm	$k_3 \neq 0 \Rightarrow \exists s \in (-1,0),\ E(s,k_3) = 0$ (IVT)	proved
E_right_neg	thm	$t \geq 1 \Rightarrow E(-10^{-4}, -t) < 0$	proved
E_left_pos	thm	$t \leq 589 \Rightarrow E(-0.9999, -t) > 0$	proved
E_left_neg	thm	$t \geq 590 \Rightarrow E(-0.9999, -t) < 0$	proved
s2	def	root selector: $E(s_2(t), -t) = 0$, $s_2(t) \in (-0.9999, -0.0001)$ for $1 \leq t \leq 589$; 0 otherwise	definition
exists_root_grid	thm	$1 \leq t \leq 589 \Rightarrow \exists s \in (-0.9999, -0.0001),\ E(s,-t) = 0$	proved
s2_spec	thm	$s_2(t) \in (-0.9999, -0.0001) \wedge E(s_2(t), -t) = 0$ $(1 \leq t \leq 589)$	proved
s2_mem	thm	$1 \leq t \leq 589 \Rightarrow s_2(t) \in (-0.9999, -0.0001)$	proved
s2_isRoot	thm	$1 \leq t \leq 589 \Rightarrow E(s_2(t), -t) = 0$	proved
abs_s2_lt_one	thm	$1 \leq t \leq 589 \Rightarrow \lvert s_2(t)\rvert < 1$	proved
FrequencyGrids/Lattice.lean — the lattice step and $N_{\mathrm{phase}} = 0$ (item 2)			
b	def	$b(t,k,\sigma) = (\arcsin(\sigma\lvert s_2(t)\rvert) - 2k\pi)/\ln 2,\ k \in \mathbb{Z}$	definition
log_two_pos	thm	$0 < \ln 2$	proved
log_two_ne_zero	thm	$\ln 2 \neq 0$	proved
b_eq_affine	thm	$b = \arcsin(\sigma\lvert s_2(t)\rvert)/\ln 2 - (2\pi/\ln 2)k$	proved
b_step	thm	$b(t,k+1,\sigma) - b(t,k,\sigma) = -2\pi/\ln 2$	proved
b_step_uniform	thm	step independent of $t, t', k, k', \sigma, \sigma'$	proved
b_strict_Anti	thm	$k \mapsto b(t,k,\sigma)$ strictly antitone	proved
b_injective	thm	$k \mapsto b(t,k,\sigma)$ injective	proved
phase_Defect	def	$\delta b_{t,k,\sigma} = b(t,k+1,\sigma) - b(t,k,\sigma) + 2\pi/\ln 2$	definition
phase_Defect_eq_zero	thm	$\delta b_{t,k,\sigma} = 0$ for all t, k, σ	proved
Nphase	def	$\#\{(t,k,\sigma) \in S : \delta b_{t,k,\sigma} \neq 0\}$	definition
Nphase_eq_zero	thm	$N_{\mathrm{phase}}(S) = 0$ for every truncation S	proved
FrequencyGrids/Uniqueness.lean — strict monotonicity, uniqueness of $s_2(t)$, certified enclosures			
rootRat	def	$f(u) = u^2/((1-u)^2(1+3u^2-u^3))$	definition
rootRat_denom_pos	thm	$u \in (0,1) \Rightarrow (1-u)^2(1+3u^2-u^3) > 0$	proved
rootRat_pos	thm	$u \in (0,1) \Rightarrow f(u) > 0$	proved
deriv_rootRat	thm	$f'(u) = u(1-u)(2+7u^3-3u^4)/((1-u)^2(1+3u^2-u^3))^2$	proved
deriv_rootRat_log	thm	$f'(u) = f(u)\,(2/(u(1-u)) - (6u-3u^2)/(1+3u^2-u^3))$	proved
logDeriv_ge_five	thm	$2/(u(1-u)) - (6u-3u^2)/(1+3u^2-u^3) \geq 5$ (via 8 and 3)	proved
rootRat_strictMonoOn	thm	f strictly increasing on $(0,1)$	proved
root_iff	thm	$E(s,-t) = 0 \iff f(s^2) = 24t^2$ on $(-1,0)$	proved
exists_unique_root_grid	thm	$\exists!\, s \in (-0.9999, -0.0001),\ E(s,-t) = 0$	proved
s2_unique	thm	any root in the bracket equals $s_2(t)$ (no dependence on 'choose')	proved
abs_s2_strictMono	thm	$t \mapsto \lvert s_2(t)\rvert$ strictly increasing	proved
s2_mem_of_sign_change	thm	sign change on $[lo, hi] \Rightarrow s_2(t) \in (lo, hi)$ (certificate)	proved
FrequencyGrids/Enclosure.lean — certified rational enclosures of $s_2(1), \ldots, s_2(10)$			
s2_enclosure_1 ... 10	thm	$s_2(t) \in (lo_t, hi_t)$, width $4 \cdot 10^{-6}$, $t = 1, \ldots, 10$	proved
FrequencyGrids/Counting.lean — counting the resonant stratum, N_R (item 3)			
natCast_toNat_eq	thm	cast bridge $\mathbb{N} \to \mathbb{Z} \to \mathbb{R}$ for toNat	proved

Declaration	Kind	Statement	Status
`card_le_of_forall_mem_Icc`	thm	$S \subseteq [a,b]$ (integers) $\Rightarrow \#S \le b-a+1$	proved
`le_card_of_forall_Icc_mem`	thm	$[a,b] \cap \mathbb{Z} \subseteq S \Rightarrow \#S \ge b-a-2$	proved
`bOf`	def	$b_\alpha(k) = (\alpha - 2k\pi)/\ln 2$	definition
`b_eq_bOf`	thm	$b(t,k,\sigma) = b_{\arcsin(\sigma\lvert s_2(t)\rvert)}(k)$	proved
`kLo`	def	$(\alpha - T\ln 2)/2\pi$	definition
`kHi`	def	$(\alpha + T\ln 2)/2\pi$	definition
`kSet`	def	$\{k \in \mathbb{Z} : \lceil k_{lo}\rceil \le k \le \lfloor k_{hi}\rfloor,\ k \ne 0\}$	definition
`kHi_sub_kLo`	thm	$k_{hi} - k_{lo} = T\ln 2/\pi$	proved
`mem_Icc_ceil_floor`	thm	$\lceil k_{lo}\rceil \le k \le \lfloor k_{hi}\rfloor \iff k_{lo} \le k \le k_{hi}$	proved
`mem_Icc_iff_abs_bOf`	thm	$k_{lo} \le k \le k_{hi} \iff \lvert b_\alpha(k)\rvert \le T$	proved
`mem_kSet_iff`	thm	$k \in k_{Set} \iff k \ne 0 \wedge \lvert b_\alpha(k)\rvert \le T$	proved
`card_kSet_bounds`	thm	$T \ge 0 \Rightarrow \lvert \#k_{Set}(\alpha,T) - T\ln 2/\pi\rvert \le 3$	proved
`sigmaSet`	def	$\{-1,1\} \subset \mathbb{R}$	definition
`card_sigmaSet`	thm	$\#\sigma_{Set} = 2$	proved
`mem_sigmaSet`	thm	$\sigma \in \{-1,1\} \iff \sigma = -1 \vee \sigma = 1$	proved
`phase`	def	$\alpha(t,\sigma) = \arcsin(\sigma\lvert s_2(t)\rvert)$	definition
`latticeSet`	def	$[1,100] \times \{-1,1\}$ (the 200 lattices)	definition
`card_Icc_1_100`	thm	$\#\{1,\dots,100\} = 100$	proved
`card_latticeSet`	thm	$\#\text{latticeSet} = 200$	proved
`modeSet`	def	triplets (t,k,σ) with $1 \le t \le 100$, $\sigma = \pm 1$, $k \ne 0$, $\lvert b\rvert \le T$	definition
`mem_modeSet`	thm	membership characterisation of the triplet set	proved
`Ntotal`	def	$N_{\mathrm{tot}}(T) = \#\text{modeSet}(T)$	definition
`NR`	def	$N_R(T) = \#\{b(t,k,\sigma) : \text{triplets}\}$	definition
`Has_Collision`	def	$\exists$ two distinct triplets with equal b	definition
`card_modeSet`	thm	$\#\text{modeSet} = \sum_{(t,\sigma)} \#k_{Set}$	proved
`Ntotal_bounds`	thm	$\lvert N_{\mathrm{tot}} - 200\,T\ln 2/\pi\rvert \le 600$	proved
`NR_le_Ntotal`	thm	$N_R(T) \le N_{\mathrm{tot}}(T)$	proved
`NR_eq_Ntotal_iff`	thm	$N_R = N_{\mathrm{tot}} \iff$ no collisions	proved
`slope_bounds`	thm	$44.12 < 200\ln 2/\pi < 44.14$	proved
`Ntotal_approx`	thm	$\lvert N_{\mathrm{tot}} - 44.13\,T\rvert \le 600 + T/100$	proved
`NR_le_slope`	thm	$N_R(T) \le 44.14\,T + 600$	proved
`card_modeSet_eq_Ntotal`	thm	$\#\text{modeSet}\,T = N_{\mathrm{tot}}(T)$	proved
`NR_eq_Ntotal_of_noCollision`	thm	no collision $\Rightarrow N_R(T) = N_{\mathrm{tot}}(T)$	proved
`FrequencyGrids/Structure.lean` — collision rigidity (item 3)			
`s2_abs_pos`	thm	$1 \le t \le 589 \Rightarrow 0 < \lvert s_2(t)\rvert$	proved
`b_collision_structure`	thm	$b(t,k,\sigma) = b(t',k',\sigma') \Rightarrow \sigma = \sigma' \wedge \lvert s_2(t)\rvert = \lvert s_2(t')\rvert \wedge k = k'$	proved
`collision_imp_duplicate_phase`	thm	a collision forces $t \ne t'$ with $\lvert s_2(t)\rvert = \lvert s_2(t')\rvert$	proved
`abs_s2_injective`	thm	$\lvert s_2(t)\rvert = \lvert s_2(t')\rvert \Rightarrow t = t'$ (same root $\Rightarrow t^2 = t'^2$)	proved
`not_hasCollision`	thm	$\neg\,\text{HasCollision}(T)$ for every T	proved
`NR_eq_Ntotal`	thm	$N_R(T) = N_{\mathrm{tot}}(T)$, unconditionally	proved
`FrequencyGrids/Resonance.lean` — Gram bound and $b \mapsto -b$ invariance (items 4, 5)			
`gramCoeff`	def	$G_{12}(T) = 1$ if $b_1 = b_2$, else $(e^{i(b_1-b_2)T} - 1)/(i(b_1-b_2)T)$	definition
`gramCoeff_self`	thm	$G_{11}(T) = 1$	proved
`norm_exp_I_sub_one_le_two`	thm	$\lVert e^{i\theta} - 1\rVert \le 2$	proved
`gramCoeff_offdiag_bound`	thm	$b_1 \ne b_2,\ T > 0 \Rightarrow \lvert G_{12}(T)\rvert \le 2/(T\lvert b_1 - b_2\rvert)$	proved
`hterm`	def	$m^{-(1/2+ib)}$ as a complex power	definition
`hsum`	def	$S_f(N,b) = \sum_{m \le N} m^{-(1/2+ib)}$	definition
`hterm_conj`	thm	$\overline{m^{-(1/2+ib)}} = m^{-(1/2-ib)}$	proved

Declaration	Kind	Statement	Status
`hsum_conj`	thm	$\overline{S_f(N,b)} = S_f(N,-b)$	proved
`hsum_re_even`	thm	$\mathrm{Re}\, S_f(N,-b) = \mathrm{Re}\, S_f(N,b)$	proved
`hsum_re_neg_iff`	thm	$\mathrm{Re}\, S_f(N,-b) = c \iff \mathrm{Re}\, S_f(N,b) = c$	proved
`FrequencyGrids/Congruence.lean` — the $6n \pm 1$ filter (item 6)			
`prime_mod_six`	thm	p prime, $p > 3 \Rightarrow p \equiv \pm 1\ (6)$	proved
`prime_zmod_six`	thm	$(p : \mathrm{ZMod}\, 6) \in \{1, 5\}$	proved
`converse_fails`	thm	$25\%6 = 1 \wedge \neg\,\mathrm{Prime}(25)$	proved
`filter_not_sufficient`	thm	$\exists n,\ n\%6 = 1 \wedge \neg\,\mathrm{Prime}(n)$	proved
`FrequencyGrids/Eigen.lean` — the Gram matrix, its spectrum, N_{off}, N_{van}			
`cpow_conj_natCast`	thm	$\overline{m^w} = m^{\bar{w}}$ for $m \in \mathbb{N}$	proved
`conj_gram_exponent`	thm	$\overline{-1 - i(x-y)} = -1 - i(y-x)$	proved
`modeB`	def	frequency b_i of a mode $i \in \mathrm{modeSet}(T)$	definition
`gramMatrix`	def	$G_{ij}(X) = \sum_{n \le X} n^{-1-i(b_i - b_j)}$	definition
`gramMatrix_isHermitian`	thm	$G(X)^{\mathsf{H}} = G(X)$, so $(G + G^{\mathsf{H}})/2 = G$	proved
`eig`	def	λ_j, the eigenvalues of $G(X)$ (spectral theorem)	definition
`eigPos`	def	$\max(\lambda_j, 0)$ — the code's clamp	definition
`eigMax`	def	$\max_j \lambda_j^+$ (as sup; $\sup \emptyset = 0$)	definition
`offSum`	def	$I_j(X) = \sum_{i \ne j} \lvert G_{ij}(X)\rvert$	definition
`offRatio`	def	$r_j(X) = I_j(X)/\lvert G_{jj}(X)\rvert$	definition
`Noff`	def	$N_{\mathrm{off}}(T, X, \delta_X) = \#\{j : r_j(X) > \delta_X\}$	definition
`Nvan`	def	$N_{\mathrm{van}}(T, X, \delta_T) = \#\{j : \lambda_j^+ \le \delta_T \max_j \lambda_j^+\}$	definition
`card_univ_modeSet`	thm	$\#\mathrm{univ} = N_{\mathrm{tot}}(T)$	proved
`Noff_le_Ntotal`	thm	$N_{\mathrm{off}} \le N_{\mathrm{tot}}$	proved
`Nvan_le_Ntotal`	thm	$N_{\mathrm{van}} \le N_{\mathrm{tot}}$	proved
`eigPos_le_eigMax`	thm	$\lambda_j^+ \le \max_i \lambda_i^+$	proved
`eigMax_nonneg`	thm	$\mathrm{modeSet}(T) \ne \emptyset \Rightarrow 0 \le \max_j \lambda_j^+$	proved
`gramE`	def	$E_{i,m} = (m+1)^{-1/2-ib_i}$, $m < \lfloor X \rfloor$	definition
`conj_half_exponent`	thm	$\overline{-1/2 - ix} = -1/2 + ix$	proved
`sum_Icc_one_eq_sum_fin`	thm	$\sum_{1 \le n \le N} = \sum_{m : \mathrm{Fin}\, N} (m+1)$	proved
`gramMatrix_eq_mul_conjTranspose`	thm	$G(X) = E\, E^{\mathsf{H}}$	proved
`gramMatrix_posSemidef`	thm	$G(X) \succeq 0$	proved
`eigenvalues_congr`	thm	eigenvalues independent of the hermitianity proof	proved
`eig_nonneg`	thm	$0 \le \lambda_j$	proved
`eigPos_eq_eig`	thm	$\max(\lambda_j, 0) = \lambda_j$ (clamp is the identity)	proved
`zero_le_eigMax`	thm	$0 \le \max_j \lambda_j^+$, also with no modes	proved
`rank_gramMatrix_le`	thm	$\mathrm{rank}\, G(X) \le \lfloor X \rfloor$	proved
`mulVec_eigenvectorBasis_eig`	thm	$Gv_j = \lambda_j v_j$ for the eigenbasis	proved
`card_nonzero_eig_eq_rank`	thm	$\#\{j : \lambda_j \ne 0\} = \mathrm{rank}\, G(X)$	proved
`card_nonzero_eig_le_rank`	thm	$\#\{j : \lambda_j \ne 0\} \le \mathrm{rank}\, G(X)$	proved
`card_le_Nvan_add`	thm	$\#\mathrm{modeSet} \le N_{\mathrm{van}} + X\ \ (\delta \ge 0)$	proved
`Ntotal_sub_le_Nvan`	thm	$N_{\mathrm{van}} \ge N_{\mathrm{tot}} - \lfloor X \rfloor\ \ (\delta \ge 0)$	proved
`NR_sub_le_Nvan`	thm	$N_{\mathrm{van}} \ge N_R - \lfloor X \rfloor\ \ (\delta \ge 0$, no hypothesis)	proved
`Noff_eq_card_of_forall_gt`	thm	$\forall j,\ \delta < r_j \Rightarrow N_{\mathrm{off}} = \#\mathrm{modeSet}$	proved
`Noff_eq_card_of_gram_gt`	thm	one neighbour with $\delta\lvert G_{jj}\rvert < \lvert G_{ij}\rvert$ for each $j \Rightarrow N_{\mathrm{off}} = \#\mathrm{modeSet}$	proved
`harmonicR`	def	$H_N = \sum_{1 \le n \le N} n^{-1}$ as a real	definition
`harmonicR_eq_harmonic`	thm	$H_N = (\mathrm{harmonic}\, N : \mathbb{Q})$	proved
`harmonicR_pos`	thm	$N > 0 \Rightarrow H_N > 0$	proved

Declaration	Kind	Statement	Status
gramMatrix_diag	thm	$G_{ii}(X) = H_{\lfloor X \rfloor}$	proved
gramMatrix_diag_eq_harmonic	thm	$G_{ii}(X)$ is Mathlib's harmonic	proved
gramMatrix_diag_pos	thm	$\lfloor X \rfloor \geq 1 \Rightarrow \lvert G_{ii} \rvert > 0$	proved
rank_gramMatrix_le_rank_gramE	thm	$\operatorname{rank} G \leq \operatorname{rank} E$	proved
card_zero_eig_ge	thm	$\#\{j : \lambda_j = 0\} \geq \#\text{modeSet} - \lfloor X \rfloor$	proved
Noff_eq_card_of_gram_gt'	thm	the same, with $\lvert G_{ii} \rvert > 0$ discharged by gramMatrix_diag_pos	proved
FrequencyGrids/Parametric.lean — $t_{\max}$ as a parameter			
latticeSetOf	def	$[1, t_{\max}] \times \{-1, 1\}$	definition
card_latticeSetOf	thm	$\#\,\texttt{latticeSetOf}\, t_{\max} = 2\, t_{\max}$	proved
modeSetOf	def	triplets for a truncation $t_{\max}$	definition
mem_modeSetOf	thm	membership: $t \leq t_{\max}$, $\sigma = \pm 1$, $k \neq 0$, $\lvert b \rvert \leq T$	proved
modeSet_eq	thm	$\texttt{modeSet}\, T = \texttt{modeSetOf}\, 100\, T$	proved
NtotalOf	def	$N_{\text{tot}}(t_{\max}, T)$	definition
NtotalOf_bounds	thm	$\lvert N_{\text{tot}} - 2 t_{\max} T \ln 2/\pi \rvert \leq 6\, t_{\max}$	proved
FrequencyGrids/GeneralGram.lean — arbitrary frequency families $G = EE^{\mathsf{H}}$, and the distinct-frequency lattice ($\#\iota = N_R$)			
gramOf	def	$G_{ij}(X) = \sum_{n \leq X} n^{-1-i(f_i - f_j)}$ for $f : \iota \to \mathbb{R}$	definition
evalOf	def	$E_{i,m} = (m+1)^{-1/2 - i f_i}$, index Fin $\lfloor X \rfloor$	definition
evalFin	def	$E_{i,m} = (m+1)^{-1/2 - i f_i}$, index Fin X $(X : \mathbb{N})$	definition
gramOf_eq_mul_conjTranspose	thm	$G = E\, E^{\mathsf{H}}$	proved
gramOf_nat_eq_mul_conjTranspose	thm	$G = E\, E^{\mathsf{H}}$ with E indexed by Fin X	proved
gramOf_isHermitian	thm	$G^{\mathsf{H}} = G$	proved
gramOf_posSemidef	thm	$G \succeq 0$	proved
gramOf_diag	thm	$G_{ii} = H_{\lfloor X \rfloor}$	proved
eigOf	def	eigenvalues λ_j of G	definition
eigOf_nonneg	thm	$0 \leq \lambda_j$	proved
rank_gramOf_le_rank_eval	thm	$\operatorname{rank} G \leq \operatorname{rank} E$	proved
rank_gramOf_le	thm	$\operatorname{rank} G \leq \lfloor X \rfloor$	proved
card_nonzero_eigOf_eq_rank	thm	$\#\{\lambda_j \neq 0\} = \operatorname{rank} G$	proved
zero_le_sSup_clamped	thm	$0 \leq \max_j \lambda_j^+$ (also for $\iota = \emptyset$)	proved
vanishOf	def	$\#\{j : \lambda_j^+ \leq \delta \max_i \lambda_i^+\}$	definition
vanishOf_ge	thm	$\#\iota \leq N_{\text{van}}(f, X, \delta) + \lfloor X \rfloor$	proved
gramMatrix_eq_gramOf	thm	$\texttt{gramMatrix}\, T\, X = \texttt{gramOf}\, (\texttt{modeB}\, T)\, X$	proved
gramMatrix_nat_eq_mul_conjTranspose	thm	$\texttt{gramMatrix}\, T\, X = E\, E^{\mathsf{H}}$ with E indexed by Fin X	proved
freqSet	def	the distinct resonant frequencies	definition
card_freqSet	thm	$\#\,\texttt{freqSet}\, T = N_R(T)$	proved
gramFreq	def	G indexed by $\texttt{freqSet}\, T$ $(\#\iota = N_R)$	definition
gramFreq_eq_mul_conjTranspose	thm	$G = E\, E^{\mathsf{H}}$	proved
gramFreq_isHermitian	thm	$G^{\mathsf{H}} = G$	proved
gramFreq_posSemidef	thm	$G \succeq 0$	proved
rank_gramFreq_le	thm	$\operatorname{rank} G \leq \lfloor X \rfloor$	proved
NvanFreq	def	N_{van} of the distinct-frequency matrix	definition
eigFreq_nonneg	thm	$0 \leq \lambda_j$	proved
NvanFreq_ge	thm	$N_R(T) \leq N_{\text{van}}(F, X, \delta) + \lfloor X \rfloor$ $(\delta \geq 0$, no hypothesis)	proved
NvanFreq_sub_ge	thm	$N_{\text{van}}(F, X, \delta) \geq N_R - \lfloor X \rfloor$	proved
FrequencyGrids/Estimate.lean — explicit lower bounds for the Gram entries			

Declaration	Kind	Statement	Status
`cos_155_ge`	thm	$\cos 1.55 \geq 1/50$	proved
`re_cpow_gram`	thm	$\mathrm{Re}(n^{-1-i\theta}) = n^{-1}\cos(\theta \log n)$	proved
`re_gramOf`	thm	$\mathrm{Re}\, G_{ij}(X) = \sum_{n \leq X} n^{-1}\cos((f_i - f_j)\log n)$	proved
`re_gramOf_ge`	thm	$\lvert f_i - f_j\rvert \log X \leq 1.55 \Rightarrow H_{\lfloor X \rfloor}/50 \leq \mathrm{Re}\, G_{ij}$	proved
`norm_gramOf_ge`	thm	$\lvert f_i - f_j\rvert \log X \leq 1.55 \Rightarrow H_{\lfloor X \rfloor}/50 \leq \lVert G_{ij}\rVert$	proved
`offRatio_ge_of_close`	thm	a close neighbour $\Rightarrow r_j \geq 1/50$	proved
`Noff_eq_card_of_close`	thm	close neighbour for each j, $\delta < 1/50 \Rightarrow N_{\mathrm{off}} = \#\mathrm{modeSet}$	proved
`close_iff_div`	thm	$\lvert b_i - b_j\rvert \log X \leq 1.55 \iff \lvert b_i - b_j\rvert \leq 1.55/\log X$	proved
`FrequencyGrids/Conjectures.lean` — stated, not proved			
`etaR`	def	$\eta_R = 1 - (N_{\mathrm{phase}} + N_{\mathrm{off}} + N_{\mathrm{van}})/N_R$	definition
`etaR_eq`	thm	$\eta_R = 1 - (N_{\mathrm{off}} + N_{\mathrm{van}})/N_R$ (uses $N_{\mathrm{phase}} = 0$)	proved
`Eta_Tends_ToOne`	def:Prop	$\eta_R(T, X(T), \delta(T)) \to 1$ for $\delta \to 0$ ($X : \mathbb{R} \to \mathbb{N}$ a parameter)	not proved
`Noff_LittleO`	def:Prop	$N_{\mathrm{off}}(T, X(T), \delta(T)) = o(N_R)$, $X : \mathbb{R} \to \mathbb{N}$ a parameter	not proved
`Nvan_LittleO`	def:Prop	$N_{\mathrm{van}}(T, X(T), \delta(T)) = o(N_R)$, $X : \mathbb{R} \to \mathbb{N}$ a parameter	not proved
`Sf`	def	S_f summed over the b-grid of sign ± 1 and $0 < \lvert k\rvert \leq K$	definition
`Resonance_Predicate`	def	$\mathrm{Re}(S_f(n-1)S_f(n+1)) < 0$	definition
`NvanFreq_LittleO`	def:Prop	$N_{\mathrm{van}}(F, T, X(T), \delta(T)) = o(N_R)$, $X : \mathbb{R} \to \mathbb{N}$ a parameter	not proved
`Primality_Criterion_Conjecture`	def:Prop	$\mathrm{Re}(S_f(n-1)S_f(n+1)) < 0 \iff n$ prime, for odd $n > 3$	not proved
`FrequencyGrids/Necessary.lean` — the growth of X forced by $N_{\mathrm{van}} = o(N_R)$			
`eventually_Ntotal_pos`	thm	$N_{\mathrm{tot}} > 0$ eventually	proved
`NvanLittleO_.eventually_le_Ntotal`	thm	$N_{\mathrm{van}} = o(N_R) \Rightarrow \forall \varepsilon > 0$, eventually $(1-\varepsilon)N_{\mathrm{tot}} \leq X$	proved
`NvanLittleO_.eventually_le_NR`	thm	same with N_R, no hypothesis on collisions	proved
`not_NvanLittleO_of_le_Ntotal`	thm	$X \leq c\, N_{\mathrm{tot}}$, $c < 1 \Rightarrow \neg N_{\mathrm{van}} = o(N_R)$	proved
`not_NvanLittleO_of_le_NR`	thm	$X \leq c\, N_R$, $c < 1 \Rightarrow \neg N_{\mathrm{van}} = o(N_R)$, no hypothesis on collisions	proved
`eventually_NR_pos`	thm	$N_R > 0$ eventually	proved
`NvanFreqLittleO_.eventually_le_NR`	thm	$N_{\mathrm{van}}(F) = o(N_R) \Rightarrow \forall \varepsilon > 0$, eventually $(1-\varepsilon)N_R \leq X$	proved
`not_NvanFreqLittleO_of_le_NR`	thm	$X \leq c\, N_R$, $c < 1 \Rightarrow \neg N_{\mathrm{van}}(F) = o(N_R)$, no collision hypothesis	proved